

SMA Sunny Central Modbus Comms Block — Register Mapping

Target: PPC_Controller [DB39] | UDT: Inverter_controller
Connection: Modbus TCP | Unit ID = 3 (grid management unit)

1. Protocol Conventions

Parameter	Value
Modbus variant	TCP/IP
Unit (Slave) ID	3
Register bit-width	32-bit values — each occupies 2 consecutive 16-bit Modbus registers
Address convention	1-based (register 1 = Modbus address 0x0000)
Write function code	FC16 — Write Multiple Registers
Read function code	FC04 — Read Input Registers (measurements) FC03 — Read Holding Registers (writable params)
Endianness	Big-endian (high word first)
Cycle	Read and write every OB30 scan (100 ms)

Scaling Formats

Format	Meaning	Example
FIX0	Integer, scaling = 1; value is in physical units	WSpt = 500 → 500 kW
FIX4	Integer × 10 000; value is a per-unit fraction	WAval = 8500 → 0.8500 pu (85 % available)

2. WRITE to SMA Inverter (PLC → Inverter, FC16)

Each OB30 cycle the comms block reads the UDT fields listed below and writes them to the corresponding SMA holding registers using FC16.

UDT Field	SMA Register Name	Reg Address	Type	Scaling	ENUM / Notes
OperMode	InvOpMod	See manual	DInt	ENUM	308 = Operation, 303 = Stop
(derived)	GriMng.RemRdy	See manual	DInt	ENUM	308 = Ready, 303 = Standby — NOT in UDT; comms block derives from OperMode
WMode	GriMng.WMod	See manual	DInt	ENUM	1079 = WCtrlCom, 303 = Off
WSpt	GriMng.WNom	See manual	DInt	FIX0 (kW)	Active power setpoint in kW (integer)
VArMode	GriMng.VArMod	See manual	DInt	ENUM	1072 = VArCtrlCom, 1075 = PFCtrlCom, 303 = Off
VArSpt	GriMng.VArNom	See manual	DInt	FIX0 (kVAr)	Reactive power setpoint in kVAr (integer)
PFSpt	GriMng.PFNom	See manual	DInt	FIX4 (×10000)	9500 = PF 0.950, 10000 = PF 1.000
ErrClr	FltRst	Holding reg 8	DInt	—	26 = Acknowledge (one-shot), 0 = idle — see one-shot rule below

Register addresses marked 'See manual' must be confirmed from:

SMA\SCADA_Modbus_Register_Map_SMA_SunnyCentral_ENUMS_Ramps_RideThrough_PPC_RO.xlsx

2a. Start / Stop Interlock Sequence

The SMA inverter enforces a strict write sequence. Writing out of order will be rejected or cause unexpected behaviour. The comms block MUST derive RemRdy internally from OperMode and guarantee the following order:

Command	Step	Register written	Value
START (OperMode → 308)	1	GriMng.RemRdy	308 (Ready)
	2	InvOpMod	308 (Operation)
STOP (OperMode → 303)	1	GriMng.RemRdy	303 (Standby)
	2	InvOpMod	303 (Stop)

2b. ErrClr One-Shot Rule

ErrClr in the UDT is set to **26** for exactly one OB30 scan (100 ms) by the FaultHandler FC, then returns to 0. The comms block must:

- Write Holding Register 8 = 26 in the scan where ErrClr ≠ 0.
- Write Holding Register 8 = 0 in all other scans.

Do NOT continuously write 26. The SMA inverter requires a rising edge (0 → 26) to acknowledge a fault — repeated writes are ignored.

3. READ from SMA Inverter (Inverter → PLC, FC04)

Each OB30 cycle the comms block reads the following SMA input registers via FC04 and writes the decoded values into the corresponding UDT fields.

UDT Field	SMA Register Name	Reg Address	Type	Scaling	Derivation / Notes
RemReady	OpStt	See manual	DInt	ENUM	TRUE when OpStt ∈ {307, 308, 309}. FALSE otherwise. See derivation table below.
Error	ErrStt	See manual	DInt	ENUM	TRUE when ErrStt ≠ 307 (307 = Ok). FALSE = no fault.
Wactive	W	See manual	DInt	FIX0 (kW)	AC active power output — measured value in kW
Qactive	VAr	See manual	DInt	FIX0 (kVAr)	AC reactive power output — measured value in kVAr
WAval	GriMng.WAval	See manual	DInt	FIX4 (pu×10000)	Available active power fraction. 10000 = 100% available.
VArAval	GriMng.VArAval	See manual	DInt	FIX4 (pu×10000)	Available reactive power fraction.
PwrOffReas	PwrOffReas	Input reg 178	DInt	ENUM / code	0 = no event. Non-zero = disconnection reason code. 21626 = Low Power SetPoint.
DrtStt	DrtStt	See manual	DInt	ENUM	0 = no derating. Non-zero = derating active (thermal, frequency, etc.)
(SCADA log)	ErrNo	Input reg 96	DInt	ENUM	Fault code — NOT stored in UDT. Log to SCADA when Error = TRUE.

3a. RemReady Derivation (from OpStt)

SMA does not expose a dedicated remote-ready bit. The comms block derives RemReady from the OpStt register:

OpStt value	Meaning	RemReady
308	Operation — producing	TRUE
309	Derating	TRUE
307	Ok (standby / ready)	TRUE
303	Stop	FALSE
35	Error	FALSE
Other	Unknown / initialising	FALSE

Verify ENUM values against the SMA Modbus manual. Adjust the table if the actual values differ.

3b. Error Derivation (from ErrStt)

ErrStt = 307 means 'Ok' (no fault). Map Error as:

```
Error := (ErrStt <> 307)
```

4. Per-Cycle Transaction Pseudocode

4a. Write Transaction (each OB30 cycle, per inverter)

May be batched into a single FC16 call. Ordering within the frame must respect the interlock sequence in Section 2a.

```
IF ErrClr <> 0 THEN
    Write FltRst (reg 8) = ErrClr // Fault acknowledge one-shot
ELSE
    Write FltRst (reg 8) = 0
END_IF

IF OperMode changed THEN
    IF OperMode = 303 THEN // Stop sequence
        Write GriMng.RemRdy = 303 // Step 1: remove remote permission
        Write InvOpMod = 303 // Step 2: stop
    ELSIF OperMode = 308 THEN // Start sequence
        Write GriMng.RemRdy = 308 // Step 1: grant remote permission
        Write InvOpMod = 308 // Step 2: start
    END_IF
END_IF

Write GriMng.WMod = WMode // Active power control mode
Write GriMng.WNom = WSpt // Active power setpoint (kW)
Write GriMng.VArMod = VArMode // Reactive control mode
Write GriMng.VArNom = VArSpt // Reactive setpoint (kVAr)
Write GriMng.PFNom = PFSpt // PF setpoint (x10000)
```

4b. Read Transaction (each OB30 cycle, per inverter)

```
Read OpStt → derive RemReady (TRUE if OpStt ∈ {307, 308, 309})
Read ErrStt → derive Error (TRUE if ErrStt <> 307)
Read W → Wactive (kW, FIX0)
Read VAr → Qactive (kVAr, FIX0)
Read GriMng.WAval → WAval (pu×10000, FIX4)
Read GriMng.VArAval → VArAval (pu×10000, FIX4)
Read PwrOffReas (reg 178) → PwrOffReas
Read DrtStt → DrtStt
// For SCADA alarm logging only (not written to UDT):
Read ErrNo (reg 96) → log when Error = TRUE
```

4c. Comms Failure Handling

```
IF read transaction fails (timeout or exception response) THEN
    Set CommError := TRUE for this inverter
    // Leave all other UDT fields at their last valid values
    // CommError cleared automatically when next successful read completes
END_IF
```

5. Local PLC Fields — NOT from Modbus

These UDT fields are managed entirely within the PLC. The comms block must NOT overwrite them.

UDT Field	Type	Source	Description
Enabled	Bool	HMI / DB39 operator input	Set TRUE to include inverter in PPC. Set by operator, not by inverter.
CommError	Bool	Comms block itself	Set TRUE if no valid Modbus response within timeout (e.g. 3 consecutive failures). Cleared when comms recover.

6. DB39 Scalar Fields Used by PPC Logic

These fields are in the DB39 PPC_Controller block. Some are set by SCADA or HMI (inputs to PPC); others are written by the PPC FBs for HMI display.

Set by SCADA / HMI (inputs to PPC)

DB39 Field	Type	Description	Source
START_CONTROLLER	Bool	Maps to EN_PPC FB input — master enable	HMI
Targets_P	Real	Active power setpoint from SCADA (kW)	SCADA (REMOTE) / HMI (LOCAL)
Targets_Q	Real	Reactive power setpoint (kVAr)	SCADA / HMI
Targets_PF	Real	Power factor setpoint	SCADA / HMI
P_RampUp	Real	Active power ramp-up rate (kW/s)	Engineering / HMI
P_RampDown	Real	Active power ramp-down rate (kW/s)	Engineering / HMI
Q_RampUp	Real	Reactive power ramp-up rate (kVAr/s)	Engineering / HMI
Q_RampDown	Real	Reactive power ramp-down rate (kVAr/s)	Engineering / HMI
WRtg_kw	Real	Rated power per inverter (kW) — read from SMA reg 184 at commissioning	Engineering
Plant_P_meas	Real	Plant-level active power at PCC (kW) — written by energy meter comms	Meter
Plant_Q_meas	Real	Plant-level reactive power at PCC (kVAr)	Meter

Written by PPC Logic (outputs for HMI / SCADA)

DB39 Field	Type	Description
Plant_Mode	Int	0 = LOCAL, 1 = REMOTE_IEC, 2 = FALLBACK
Plant_N_Online	Int	Number of inverters currently online
Limits_PmaxPlant	Real	Total available active power (kW)
Limits_QmaxPlant	Real	Total available reactive power (kVAr)
Ramps_Pcmd	Real	Rate-limited active power command (kW)
Ramps_Qcmd	Real	Rate-limited reactive power command (kVAr)
AnyFault	Bool	TRUE = at least one inverter fault, comm error, or PwrOffReas ≠ 0
AnyDerating	Bool	TRUE = at least one healthy inverter is derated
FaultMask	Word	Bit i = inverter i has active fault or comm error

7. Inverter_controller UDT — Complete Field Reference

All fields of the Inverter_controller UDT and their origin. The comms block reads/writes only the Modbus-mapped fields; all others must be left untouched.

Field	Type	Direction	Origin
Enabled	Bool	PLC → UDT	HMI operator input — include in PPC
RemReady	Bool	Modbus → UDT	Derived from OpStt register (FC04)
Error	Bool	Modbus → UDT	Derived from ErrStt register (FC04)
CommError	Bool	Comms block	Set by comms block on timeout/failure

Field	Type	Direction	Origin
OperMode	DInt	UDT → Modbus	Written by FaultHandler FC; maps to InvOpMod + RemRdy (FC16)
WMode	DInt	UDT → Modbus	Written by PowerDistribution FC; maps to GriMng.WMod (FC16)
WSpt	DInt	UDT → Modbus	Written by PowerDistribution FC; maps to GriMng.WNom (FC16)
VArMode	DInt	UDT → Modbus	Written by ReactiveControl FC; maps to GriMng.VArMod (FC16)
VArSpt	DInt	UDT → Modbus	Written by ReactiveControl FC; maps to GriMng.VArNom (FC16)
PFSpt	DInt	UDT → Modbus	Written by ReactiveControl FC; maps to GriMng.PFNom (FC16)
ErrClr	DInt	UDT → Modbus	Written by FaultHandler FC; maps to FltRst reg 8 (FC16) — one-shot
Wactive	DInt	Modbus → UDT	AC active power output; maps to W register (FC04)
Qactive	DInt	Modbus → UDT	AC reactive output; maps to VAr register (FC04)
WAval	DInt	Modbus → UDT	Available active power fraction (FIX4); maps to GriMng.WAval (FC04)
VArAval	DInt	Modbus → UDT	Available reactive fraction (FIX4); maps to GriMng.VArAval (FC04)
PwrOffReas	DInt	Modbus → UDT	Disconnection reason code; maps to reg 178 (FC04). 0 = normal.
DrtStt	DInt	Modbus → UDT	Derating status; 0 = none active. FC04.

ErrClr : DInt must be added to the Inverter_controller UDT in TIA Portal if not already present.

Register addresses marked 'See manual' must be confirmed from:

SMA\SCADA_Modbus_Register_Map_SMA_SunnyCentral_ENUMS_Ramps_RideThrough_PPC_RO.xlsx

Document generated from PPC_ENERGY project. For SCL source code see FB_PPC_Controller.scl and associated FCs.